3. Literature Review & Evaluation

Project: NutriVision — AI-Driven Dietary Health Assistant
Document status: Initial review draft
Note: This document follows the assignment brief. The instructor's direct feedback and the official grading weights will be added when received.

3.1 Problem Background

Diet has a direct relationship with health and well-being, but healthy eating is not one identical diet for everyone. The World Health Organization describes healthy diets using principles such as adequacy, balance, moderation, and diversity, while also noting that individual needs, cultural context, and local food availability affect the final choices. This supports the need for guidance that is understandable and adaptable rather than a one-size-fits-all meal list.

Users commonly face three related problems:

1. Nutritional information can be difficult to interpret during everyday decisions.
2. Existing tracking methods may require repetitive manual entry without useful feedback.
3. Generic recommendations may not reflect dietary restrictions, preferences, or local habits.

NutriVision addresses these problems by combining meal logging, nutritional insight, alternative suggestions, and personalized recipes in one cross-platform experience.

3.2 Existing Solution Categories

Category	Strengths	Limitations	NutriVision opportunity
Manual food diaries	Simple and transparent	Often provide little analysis or feedback	Add useful summaries while keeping the log understandable
Nutrition databases and trackers	Can provide detailed nutrient information	May be complex and require accurate portion input	Present the most relevant information in a simpler flow
Recipe applications	Offer many recipes and preparation ideas	Recommendations may not reflect personal restrictions	Filter suggestions by preferences and restrictions
Generic AI assistants	Natural-language interaction and flexible suggestions	Outputs can be incomplete, inaccurate, or unsafe if not constrained	Validate responses, show limitations, and keep the scope informational

This comparison is a design-oriented review, not a claim that NutriVision is clinically validated. The team should add peer-reviewed studies and a formal comparison of selected applications before the final report.

3.3 Evidence and Technical Basis

- WHO and FAO guidance supports using varied, balanced, moderate, and culturally appropriate dietary guidance as the conceptual basis for the product.
- AI-generated content should be treated as an assistive output that requires clear boundaries, validation, and user-facing disclaimers.
- Cloud Firestore is suitable for the planned synchronization use case because it supports a document data model, real-time listeners, offline support, and security rules. The final implementation must still configure authentication and least-privilege rules correctly.
- A cross-platform Flutter interface is appropriate for the project because the required experience is primarily mobile UI, networking, local state, and cloud synchronization.

3.4 Project Evaluation

Strengths

- The problem is relevant to everyday health and wellness decisions.

- The feature set combines practical guidance with an understandable user flow.
- The planned use of responsive UI, networking, Firestore, and automated tests demonstrates the required cross-platform skills.
- The project can be evaluated with measurable user-flow and technical criteria.

Current limitations

- Nutrition and AI accuracy depend on the selected external data/service provider.
- The application must avoid medical claims and must clearly communicate uncertainty.
- Real-time synchronization requires careful authentication and Firestore rule design.
- The MVP scope may be too broad if image recognition, advanced charts, and multiple integrations are implemented at the same time.
- The project currently needs validated user feedback and a finalized data model.

Evaluation method

The team will evaluate the MVP using:

1. Functional test cases for meal entry, analysis, recipes, alternatives, history, and synchronization.
2. Unit tests for parsing, validation, and nutritional calculations.
3. Widget tests for loading, empty, success, error, and responsive UI states.
4. Usability scenarios completed by representative test users.
5. Security checks for secret handling and unauthorized Firestore access.
6. A final demonstration mapped to the approved requirements and milestones.

3.5 Suggested Improvements

Priority	Improvement	Reason	Timing
P0	Freeze and document the MVP scope	Prevents late feature expansion	Before implementation
P0	Add response validation and safe error handling	Reduces the risk of displaying unsafe or unusable AI output	During integration
P0	Define authentication and Firestore Security Rules	Protects user meal records	Before cloud data testing
P0	Add a visible health-information disclaimer	Sets safe user expectations	First user-flow design
P1	Test with realistic local meals and restrictions	Improves usefulness and cultural relevance	During usability testing
P1	Add offline/read-only history behavior	Helps users when connectivity is unstable	After core sync
P1	Add accessibility and responsive-layout checks	Improves usability across devices	During UI implementation
P2	Consider barcode or image recognition	Could reduce manual input, but increases complexity	Future enhancement
P2	Add richer trend analysis	Could improve long-term engagement	After MVP validation

3.6 Feedback and Evaluation Record

This section is prepared for the instructor's assessment. It must be updated after each review instead of presenting assumptions as confirmed feedback.

Review date	Reviewer	Feedback	Decision/action	Status
TBD	Rana Mohy	TBD	TBD	Pending
TBD	Team review	Confirm MVP scope, roles, and data source	Update planning and requirements documents	Pending

3.7 Final Grading Criteria Mapping

The assignment brief mentions documentation, implementation, testing, and presentation. Exact marks and weighting are TBD until the official rubric is provided.

Evaluation area	Evidence prepared by NutriVision	Status
Documentation	Proposal, plan, requirements, literature review, README, architecture notes	In progress
Planning and management	Timeline, milestones, roles, risks, KPIs, decision log	Initial draft
Requirements quality	Stakeholders, personas, user stories, use cases, functional/non-functional requirements	Initial draft
Implementation	Working Flutter MVP, networking, Firestore, responsive UI	Pending development
Testing	Unit tests, widget tests, integration/acceptance scenarios, bug report	Pending development
Security and privacy	Secret handling, runtime permission review, authentication, Firestore rules	Pending design/integration
Presentation	Demo flow, screenshots, slides, contribution summary, limitations	Pending finalization

3.8 References

1. World Health Organization — Healthy diet
2. WHO and FAO — What are healthy diets?
3. Firebase — Cloud Firestore documentation
4. Flutter documentation

The final academic report should add peer-reviewed research papers and a formal comparison of comparable dietary-health applications after the team confirms the required citation style.